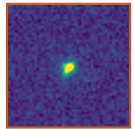
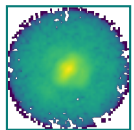


galaxy



PSF



Joint D_4 orbit

1

$\hat{R}_{90}$

$\hat{R}_{90}^2$

$\hat{R}_{90}^3$

$\hat{P}$

$\hat{P}\hat{R}_{90}$

$\hat{P}\hat{R}_{90}^2$

$\hat{P}\hat{R}_{90}^3$

galaxy and PSF transformed jointly

1. Inputs

53×53 pixels

2. Joint orbit

8 joint variants

galaxy
CNN

PSF
CNN

fusion

F

*one parameter set
for all 8 variants*

$\phi_0, \dots, \phi_7$

3. Shared F

shared weights

Features ϕ_i

$8 \times 13^2 \times 64$

4. Reynolds

signed / invariant

inverse-align
 $\psi_i^{-1}\phi_i$

$$\Psi_{1,2} = \frac{1}{8} \sum_i w_i \psi_i^{-1} \phi_i$$

$$\frac{1}{8} \sum_i \psi_i^{-1} \phi_i$$

signed / sign-free
averages

Ψ_1

Ψ_2

scalar

Ψ maps

$3 \times 13^2 \times 64$

Attention maps

maps 1-4



from scalar map

5. Pooling

4 spatial maps

bias-free
odd MLP

e'_1, e'_2

invariant
MLP

size, flux

6. Output heads

shape + scalars